From Static Traces to Pseudodynamic Handwriting Minimap Signatures

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Abstract: A digitizing tablet records the movement behind a handwritten trace, but an ordinary image retains only its shape. We study how much of that additional information can be carried by colour in a compact handwriting image. Starting with a static reconstruction, we encode speed, pressure, altitude and azimuth separately, then compare all four three-signal combinations using the same texture descriptor and support vector machine. The experiments cover 597 recordings from 75 PaHaW participants across eight tasks, with participant-separated repeated nested model fitting. Speed, altitude and azimuth give the highest task-mean macro F1, 0.5752, compared with 0.5643 for static; the difference is not supported after multiplicity correction. In an exploratory comparison on this SVM-selected encoding, SVM, logistic regression and Random Forest exceed a small CNN trained from scratch in task-mean macro F1. Combining tasks yields macro F1 of 0.6498 for logistic regression and 0.6488 for SVM. Grad-CAM concentrates on handwriting, yet agrees poorly with occlusion. Under the tested normalization and evaluation, the added colour signals provide no clear advantage over static geometry.

1 INTRODUCTION

A handwritten line shows where the pen travelled. It does not show how quickly the pen moved, how firmly it pressed, or how its inclination changed. A digitizing tablet records these quantities together with the planar trajectory. The representation problem is to make them available to an image classifier without losing the recognizable structure of the writing.

We start with a monochrome reconstruction of the visible trace, then vary one colour channel using one recorded signal. After measuring these isolated additions, we combine signals into RGB images. Each stage asks whether the additional information improves classification under the same evaluation procedure.

The application is the distinction between Parkinson's disease (PD) and healthy controls (H) in PaHaW, a database of online handwriting and drawing tasks (Drotár et al., 2016). Handwriting exposes aspects of fine motor behaviour, but the relation between pen motion and diagnosis also depends on the task, device and participant characteristics (Impedovo and Pirlo, 2019).

We call the images *pseudodynamic handwriting minimap signatures*: compact visual representations of task executions, derived from dynamic recordings. The term minimap follows work using compact source-code renderings as visual signatures (Gebara Jr. et al., 2025). Here, planar position supplies the image coordinates and colour carries measured pen attributes. A signature denotes this representation, not a person's authentication signature.

Handwriting classification has often used explicitly designed kinematic and pressure features. Drotár et al. compared such features with several classifiers on PaHaW (Drotár et al., 2016). Learned features provide another route, either from pen dynamics (Pereira et al., 2016) or from visual attributes of the reconstructed writing (Moetesum et al., 2019). These studies motivate evaluating both a fixed image descriptor and a network that learns from pixels. They do not imply that a learned representation will be preferable with only a few dozen participants per task.

The closest representation precedent is the dynamically enhanced handwriting of Diaz et al. (Diaz et al., 2019). Their images encode movement through acquired points and pen-up trajectories. Our experiment instead assigns named signals to colour channels while holding the visible trajectory fixed. This gives each added measurement a defined spatial and chromatic role that can be tested against the same static reconstruction.

Our contribution is an incremental evaluation of explicit colour encoding: a static baseline, four isolated signals and four RGB combinations under identical participant splits, followed by classifier comparison, task fusion and CNN explanation.

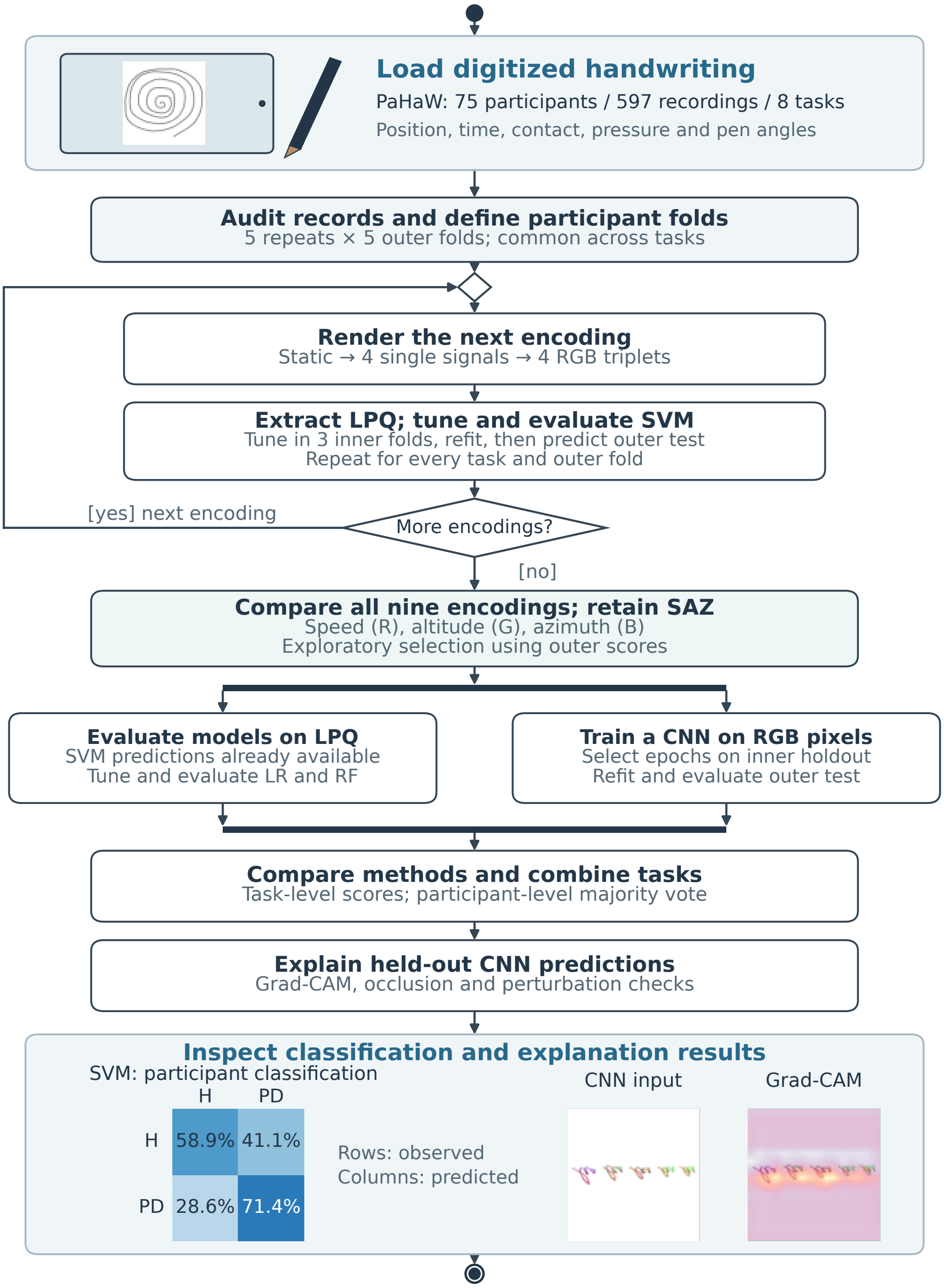

Figure 1: Illustrated activity diagram of the incremental study. Nine encodings are evaluated with LPQ and SVM using participant-separated outer folds. Selecting *SAZ* from the outer scores is exploratory. The classifier branches reuse the participant assignments; LR denotes logistic regression and RF random forest. Task scores and participant voting are distinct endpoints. XAI uses the CNN checkpoints responsible for the held-out predictions. The final illustrations show an aggregate SVM confusion matrix and one previously selected CNN explanation, from different analyses. The tablet and stylus are schematic; the handwriting and results are from the recorded experiments.

The colour encoding carried forward was selected from its cross-validated results on this cohort. The downstream comparison is consequently exploratory, with no independent test of representation selection.

Figure 1 connects the representation experiments to the classifier and XAI analyses. The loop isolates the encoding comparison; the two later branches reuse its participant partitions.

2 BUILDING THE REPRESENTATION

2.1 The Visible Trace

Let a recording contain samples

$$q_i = (x_i, y_i, t_i, s_i, z_i, a_i, p_i), \tag{1}$$

where s_i indicates surface contact, z_i is azimuth, a_i is altitude and p_i is pressure. The stored PaHaW columns place y before x; the renderer restores their planar order.

The static image uses only position and the contact indicator. A segment between samples i and $i+1$ is drawn if both samples are on the surface:

$$\mathcal{S} = \{i : s_i = 1 \text{ and } s_{i+1} = 1\}. \tag{2}$$

This preserves pen lifts as gaps instead of joining the end of one stroke to the beginning of the next. Neither the in-air trajectory nor the duration of a lift appears in the image. Contact points alone determine its bounding box, so an in-air excursion cannot change the scale of the visible writing.

The bounding box is centred on a 512×512 white canvas with a 20-pixel margin, using one scale factor for both axes. Segments are drawn in RGB $(30, 30, 30)$ with constant 4-pixel thickness. Rendering takes place at three times the target resolution and is reduced with Lanczos filtering. These settings remain fixed throughout the colour experiments.

The image retains form and stroke discontinuities, but fitting every record to its own canvas removes absolute size. Geometrically similar samples of different sizes can produce the same image, discarding information about absolute micrographia.

2.2 Adding One Signal

Four measurements are candidates for colour: speed (S), pressure (P), altitude (A) and azimuth (Z). Pressure is the tablet's contact-pressure signal. Altitude describes pen inclination relative to the surface, while azimuth describes its orientation in the tablet plane. We first encode each signal alone, always in the red channel, with green and blue fixed at 30. This avoids changing the channel assignment while screening the signals.

For a surface segment, speed is calculated directly from the consecutive coordinates and timestamps:

$$v_i = \frac{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}}{t_{i+1} - t_i}. \tag{3}$$

We use $\log(1 + v_i)$ in the tablet's native coordinate and time units. This compresses large values before the intensity mapping. Pressure and altitude use the arithmetic mean of the two endpoint measurements.

Azimuth requires a circular reference. We convert the recorded tenths of a degree to radians and compute the mean direction over contact samples. Each angle is centred on that direction and wrapped into $[-\pi, \pi]$ before the endpoint average is taken. This reduces the effect of an arbitrary zero-degree boundary. It is a circular centring operation, not temporal unwrapping, and it does not remove every possible branch-cut artefact.

Each transformed signal is scaled within the recording being rendered. Let $q_{.05}$ and $q_{.95}$ be its fifth and ninety-fifth percentiles over surface segments. The normalized value is

$$u_i = \operatorname{clip}\left(\frac{f_i - q_{.05}}{q_{.95} - q_{.05}}, 0, 1\right), \tag{4}$$

with $u_i = 0.5$ if the percentile range is degenerate. We map it to an integer channel value

$$c_i = \operatorname{round}(30 + 190 u_i). \tag{5}$$

The white background stays outside the semantic range. Antialiasing introduces intermediate intensities at boundaries, as in the static reconstruction.

This normalization needs neither a cohort-wide reference distribution nor a class label. A future record can be rendered independently. The cost is that colour describes relative variation within that recording. The same red value in two images does not imply the same absolute speed or pressure. Absolute pen orientation is also removed by circular centring. These losses matter when interpreting the final comparison with classifiers that use physical measurements directly.

2.3 Combining Signals in RGB

Three channels cannot carry four independent scalar signals without an additional design choice. We evaluate the four subsets of size three, rather than deciding in advance which measurement to discard. Table 1 gives the assignments, and Figure 2 illustrates

Figure 2: Building a pseudodynamic image from the same spiral trace. Rows show one control and one PD example, selected by a fixed identifier rule without reference to prediction quality. Columns move from static geometry to speed alone and two RGB triplets. The trajectory is unchanged; colour describes within-recording variation.

Table 1: Incremental colour encodings. A dash denotes a constant channel value of 30. *S*: speed; *P*: pressure; *A*: altitude; *Z*: azimuth.

Encoding	Red	Green	Blue
Static	–	–	–
One signal	*S*, *P*, *A* or *Z*	–	–
SPA	*S*	*P*	*A*
SPZ	*S*	*P*	*Z*
SAZ	*S*	*A*	*Z*
PAZ	*P*	*A*	*Z*

Table 2: Task inventory. T1 contains 36 PD and 36 H records; each other task contains 37 PD and 38 H records.

ID	Task	Records
T1	Archimedean spiral	72
T2	Repeated *l*	75
T3	Repeated *le*	75
T4	Repeated *les*	75
T5	Repeated *lektorka*	75
T6	Repeated *porovnat*	75
T7	Repeated *nepopadnout*	75
T8	*Tramvaj dnes už nepojede.*	75

the progression. The static image and the isolated additions use the same geometry, background and rendering settings as these combinations.

The comparison holds the classifier procedure fixed. Because its descriptor concatenates the same histogram construction for each channel, permuting the RGB channels only permutes three feature blocks. Standardization followed by a linear or radial-basis SVM is invariant to that common permutation, apart from numerical effects. The RGB assignments thus specify a reproducible representation without requiring six permutations of every triplet for the SVM experiment. This argument does not extend to every image model; the later CNN receives one fixed RGB assignment.

Segments are drawn in acquisition order. A later segment can overwrite an earlier colour at a crossing, and the image has no separate layer for repeated visits to the same position. It preserves spatially located signal variation, not the complete sample sequence. Calling the result pseudodynamic reflects that distinction.

3 DATA AND EVALUATION

3.1 Participants and Tasks

PaHaW provides eight tasks recorded with a Wacom Intuos 4M tablet (Drotár et al., 2016). We calculate speed from the recorded timestamps rather than an assumed sampling interval. Our local inventory contains 597 recordings from 75 participants: 37 with PD and 38 controls. Table 2 specifies the task set. Three spiral recordings are absent from the local corpus; the other tasks have one recording for every participant. We retain all available records and do not interpret missing files as quality exclusions.

An input audit checked file counts, sample counts, finite values and timestamp increments. Five files have a declared sample count different from the number of rows present. The renderer uses the rows actually available. There are 253 negative timestamp increments in 151 recordings, but all 1,065,788 segments with contact at both endpoints have positive increments. Equation 3 therefore requires no interpolation or imputation for the segments used in this experiment. None of the signal normalization ranges was degenerate.

The participant is the independent observational unit; tasks and alternative renderings do not enlarge the cohort. Mean age is 69.32 years in PD and 62.42 in H. Age and clinical fields are excluded from classifier inputs, but handwriting may still carry this confounding.

3.2 A Common Partition for Every Task

We construct stratified folds from one row per participant, then propagate the assignments to every task and rendering (Figure 3). Five repetitions of five outer folds produce a test prediction for every available record in each repetition. All tasks from a participant are in the same outer fold. The spiral task uses the participants for whom a record exists.

Within each outer training partition, three participant-separated inner folds choose the classical model hyperparameters. Feature standardization is fitted on the corresponding training portion, never on its validation or test participants. The selected model is refitted on all outer training participants and evaluated once on the outer test set. The rendering is independent per record, so it has no parameters learned from the other records before splitting.

Each fixed encoding and classifier yields 200 outer models and 2,985 outer-test predictions. These are repeated predictions of 597 records from 75 participants. Every later stage reuses the saved assignments.

The highest-ranked encoding was selected from these outer scores for subsequent comparisons. Nested fitting protects model hyperparameter estimation, but not that representation choice. Earlier explorations also used PaHaW. The downstream results thus concern a fixed, selected encoding on a development cohort, without an independent estimate of the complete selection procedure.

3.3 Metrics and Paired Comparisons

Macro F1 gives equal weight to the two diagnosis classes:

$$F_{\text{macro}} = \frac{F_{1,\text{H}} + F_{1,\text{PD}}}{2}. \tag{6}$$

For each task and repetition, we concatenate the five outer-test folds and compute the metric once. We then average the five resulting scores. For the representation and classifier rankings, the endpoint is the unweighted mean across all eight tasks:

$$\overline{F}_{\text{tasks}} = \frac{1}{8 \cdot 5} \sum_{t=1}^{8} \sum_{r=1}^{5} F_{t,r}. \tag{7}$$

We also retain accuracy, balanced accuracy, PD sensitivity, H specificity, positive-class F1 and ROC AUC. A mean of task scores answers a different question from a single diagnosis obtained by combining tasks; Section 6 defines the latter.

Confidence intervals use 10,000 diagnosis-stratified bootstrap draws of participants. A draw preserves all available tasks and repeated predictions for each selected person. For a paired contrast, the same draw is applied to both methods. The randomization test exchanges method predictions per participant, jointly across tasks and repetitions, using 10,000 participant-wise swap patterns. It tests the difference between fixed sets of predictions, not a shuffled-label learning pipeline.

Holm adjustment covers 36 encoding pairs; two separate 32-contrast task families for isolated signals and triplets against static; six classifier pairs at each endpoint; and eight task-specific CNN–SVM contrasts. Correction is within each declared family, not across all decisions in the broader research history. The intervals are pointwise, not simultaneous.

Both resampling procedures leave models fixed. Intervals and tests describe conditional uncertainty in these predictions, without the full variability of retraining or selection.

4 FROM SHAPE TO COLOUR

4.1 A Fixed Descriptor for the Comparison

We compare the nine encodings with one image descriptor and the same SVM search. Local Phase Quantization (LPQ) was introduced for texture classification using local Fourier phase information (Ojansivu and Heikkilä, 2008). We use a fixed channel-wise implementation, so a change in the result can be associated with a change in the rendering under the same extraction procedure.

Before extraction, foreground is detected by any channel intensity below 250. We crop its bounding box, add a margin equal to 5% of the longer dimension, pad to a square with white, and resize to 64×64 by bilinear interpolation. LPQ filters each channel with four complex 5×5 kernels at frequency offsets $(1,0)$, $(0,1)$, $(1,1)$ and $(1,-1)$, scaled by $1/5$. The signs of their real and imaginary responses form an

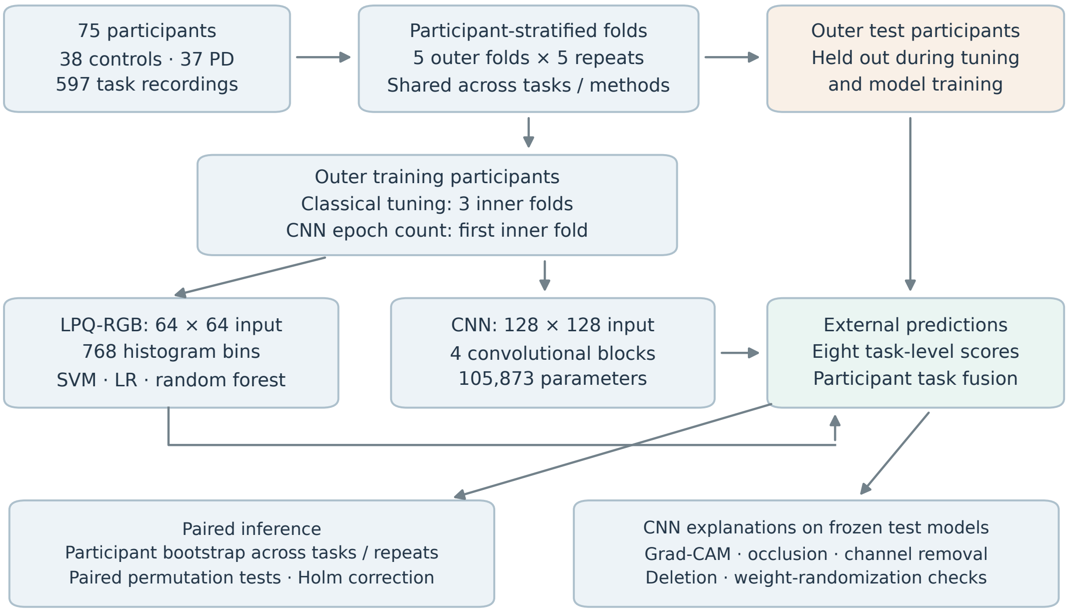

Figure 3: Participant separation throughout model fitting and evaluation. The CNN selects its epoch count on the first inner holdout; the classical models average three inner-fold scores. The outer assignments are common across tasks and methods. This diagram describes model fitting for a fixed encoding; representation selection from the outer scores remains exploratory.

eight-bit code at each pixel, using reflected boundaries. We normalize a 256-bin histogram by its count, take its element-wise square root, and concatenate the channels. The 768-attribute vector includes the padded background. No decorrelation or spatial pyramid is applied.

The static RGB channels are identical, so the static vector repeats the same histogram three times. This keeps the interface and nominal dimension constant; it does not create additional static information. For the isolated signals, two channels remain constant along the trace and retain its geometric pattern.

Standardization and SVM fitting occur inside each training partition. The search contains five linear models with $C \in \{0.01, 0.1, 1, 10, 100\}$ and sixteen radial-basis models with $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\text{scale}, 10^{-4}, 10^{-3}, 10^{-2}\}$. We use inverse-frequency class weights. The candidate with the highest mean inner-fold macro F1 is chosen, with ties resolved by the stored candidate order. The SVM learns from image texture; it never receives the original signal measurements as tabular features (Cortes and Vapnik, 1995).

4.2 The Static Baseline

The static reconstruction reaches a task-mean macro F1 of 0.5643. Its best task is repeated *les* (T4), with 0.6502 and a participant-bootstrap interval of [0.5582, 0.7351]. The spiral reaches 0.5915, while T5 is close to 0.5 at 0.5038. Static performance varies across tasks.

All tasks remain in the subsequent comparison; the strongest static task is not selected as a new benchmark.

4.3 Isolated Signals and RGB Combinations

Figure 4 orders the encodings by Equation 7. Altitude is the strongest isolated addition at 0.5718, followed by azimuth at 0.5688. Speed alone reaches 0.5625 and pressure 0.5568. Neither the presence of a dynamic signal nor an intuitively relevant physiological measurement guarantees a better image descriptor.

The individual-task contrasts vary in direction. For example, speed changes the spiral score from 0.5915 to 0.6153 but changes repeated *les* from 0.6502 to 0.5994. Altitude increases the spiral score to 0.6294 and changes T4 only slightly, to 0.6548. None of the 32 isolated-signal versus static contrasts survives its Holm correction.

Combining signals produces a similarly narrow ranking. Starting from *PAZ*, replacing pressure, azimuth or altitude with speed gives *SAZ*, *SPA* or *SPZ*, respectively. *SAZ* leads at 0.5752, only 0.0006 above *PAZ*; *SPA* reaches 0.5697 and *SPZ* reaches 0.5574. These results provide no basis for declaring pressure

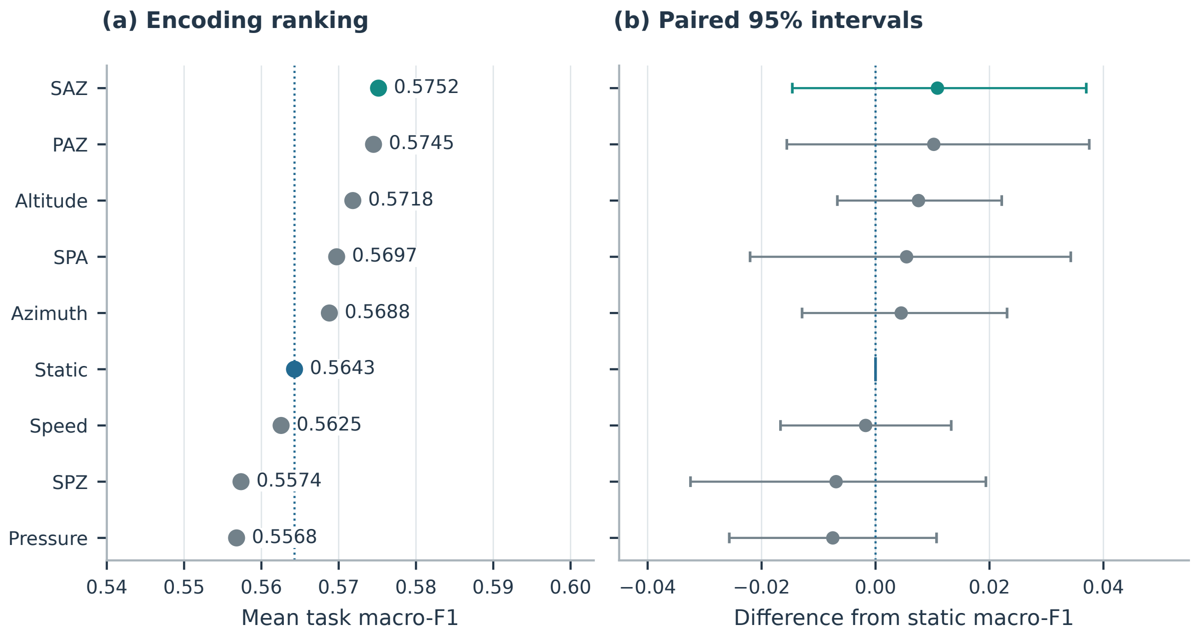

Figure 4: Encoding comparison with SVM. Left: unweighted mean of task-specific macro F1 over eight tasks and five repetitions. Right: paired differences from static with conditional 95% participant-bootstrap intervals. The complete family of 36 pairwise comparisons supports no difference after Holm adjustment.

intrinsically uninformative. LPQ summarizes normalized image patterns, not the physical measurements directly.

The paired *SAZ*–static difference is +0.0109 with a 95% interval of [−0.0146, 0.0370]. Its randomization p is 0.4096 and Holm-adjusted p is 1.0000 across the 36 encoding pairs. The *SAZ*–*PAZ* interval is [−0.0185, 0.0198], also with adjusted $p = 1.0000$. The numerical leader is therefore a practical choice for the next experiment, not a demonstrated improvement over the simpler reconstruction.

We retain *SAZ* for the remaining comparisons. Its per-task changes help interpret the modest average: the spiral improves by 0.0802, T6 by 0.0524, and T7 by 0.0455, whereas T2 falls by 0.0612. None of its eight task-specific contrasts against static remains significant within the 32-comparison triplet family.

5 FROM TEXTURE TO LEARNED FEATURES

5.1 Linear and Tree-Based Alternatives

With the *SAZ* image fixed, we compare SVM with regularized logistic regression and Random Forest. All three consume the same 768 LPQ attributes. Logistic regression is a useful linear reference because it can succeed without fitting a nonlinear decision boundary. We search L_1 and L_2 penalties with $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$, using the liblinear solver and inverse-frequency class weights. Standardization remains inside the training pipeline.

Random Forest supplies a different form of nonlinear modelling through an ensemble of decision trees (Breiman, 2001). Each candidate uses 100 trees, the Gini criterion and class weights calculated within bootstrap samples. The search combines maximum-feature choices $\{\sqrt{d}, 0.25d\}$ with minimum leaf sizes $\{1, 2, 4\}$, where $d = 768$. Tree depth is unrestricted. Both algorithms reuse the same threefold inner search and the same outer test assignments as SVM.

5.2 Learning Directly from Pixels

The CNN removes the fixed texture descriptor. Convolutional learning has a long history in document recognition (LeCun et al., 1998), but the number of independent participants here is small. We use a compact network trained from scratch to establish a direct pixel-learning baseline; this experiment is not a search across modern pretrained architectures.

The input follows the same foreground crop and square padding as LPQ but is resized to 128×128. Pixel values are mapped from $[0, 255]$ to $[-1, 1]$ with fixed constants. Four blocks contain a bias-free 3×3 convolution, batch normalization, ReLU and 2×2

max pooling. Their channel counts are 16, 32, 64 and 128. A dropout layer with probability 0.3 and one linear output act on the flattened $128 \times 8 \times 8$ feature map. The network has 105,873 trainable parameters.

Training uses AdamW with learning rate 10^{-3}, weight decay 10^{-4}, batch size 16 and binary cross-entropy with positive weight equal to the H/PD training-count ratio. Training augmentation uses rotations up to 3°, translations up to 3% and scale changes between 0.95 and 1.05. White fills exposed pixels; writing is never flipped.

For each outer fold and task, the first of the saved inner splits chooses the number of epochs. Selection uses validation macro F1, with binary cross-entropy as the tie-breaker. Training runs for at most 60 epochs, with a patience of ten and stopping enabled after epoch twelve. The selected epoch itself may be earlier than twelve. We then reinitialize the network and train on all outer training participants for exactly that number of epochs. The outer test set is used only for the saved evaluation. Unlike the classical grid searches, the CNN epoch selection uses one inner holdout rather than an average over all three inner folds.

Selected counts range from 1 to 48 epochs, with median 15. All 200 outer checkpoints are retained for XAI.

5.3 Task-Level Comparison

SVM leads the task-mean ranking at 0.5752, followed by logistic regression at 0.5674 and Random Forest at 0.5599. Their differences are small relative to the paired uncertainty. SVM minus logistic regression is 0.0077, with interval [−0.0074, 0.0230] and Holm-adjusted $p = 0.6563$. SVM minus Random Forest is 0.0152, with interval [−0.0066, 0.0365] and adjusted $p = 0.5645$.

The CNN reaches 0.5105. The SVM–CNN contrast is 0.0646, with interval [0.0277, 0.1021] and adjusted $p = 0.0054$ in the six-pair classifier family. Random Forest and logistic regression also exceed the CNN in that analysis, with adjusted $p = 0.0460$ and $p = 0.0255$, respectively. These are conditional comparisons of the saved cross-validated predictions as defined in Section 3.

The task pattern in Figure 5 differs from the overall ranking. Random Forest is strongest on T4 at 0.6791, even though its eight-task mean is below SVM. The CNN is slightly above SVM on T2 and T5, two tasks on which SVM itself performs poorly. Its task-specific differences against SVM do not survive the family of eight contrasts.

The experiment favours LPQ-based models under these choices. Architecture is not the only difference: LPQ uses 64×64 inputs, the CNN uses 128×128, and their selection budgets differ. Pretrained features, as used by Diaz et al. (Diaz et al., 2019), remain a distinct approach.

6 COMBINING HANDWRITING TASKS

A participant can be represented by several task executions rather than one. Late fusion combines their predictions while keeping each task model separate. Because all tasks share the same participant folds, the models used for a held-out person's fusion were all fitted without that person.

We first considered three SVM rules over every available task: the mean calibrated PD probability, a reliability-weighted mean, and majority voting. Sigmoid calibration uses the three inner participant folds with ensemble averaging. Reliability is derived from inner-validation macro F1, with weight $\max(F_{\text{inner}} - 0.5, 0.01)$. We do not select a subset of tasks according to outer-test performance. A missing spiral is omitted from that participant's denominator.

The initially declared probability-mean endpoint reaches macro F1 0.6341, with interval [0.5417, 0.7189]. Reliability weighting reaches 0.6447. Majority voting has the largest observed score, 0.6488, but the three rules do not differ significantly after correction. Majority minus probability mean is 0.0147, with interval [−0.0317, 0.0618] and adjusted $p = 1.0000$.

We use majority voting as a common descriptive participant endpoint for all four classifiers. If $\mathcal{T}_j$ is the set of tasks available for participant j, its score in repetition r is

$$s_{j,r} = \frac{1}{|\mathcal{T}_j|} \sum_{t \in \mathcal{T}_j} \widehat{y}_{j,t,r}, \qquad \widehat{y}_{j,r} = \mathbf{1}[s_{j,r} \geq 0.5]. \tag{8}$$

Tied votes are assigned to PD by this fixed threshold. We compute the participant-level metric separately for each repetition and average the five scores. We do not average probabilities across repetitions to create another ensemble.

Figure 6 places logistic regression at 0.6498 and SVM at 0.6488 after fusion, both with mean accuracy 0.6507. Random Forest reaches macro F1 0.6259 and the CNN 0.5544. Logistic regression's macro F1 exceeds SVM by only 0.0010, with interval [−0.0398, 0.0403] and adjusted $p = 1.0000$. No pairwise fusion contrast remains significant after Holm correction. Failure to detect a difference is not a formal equivalence result.

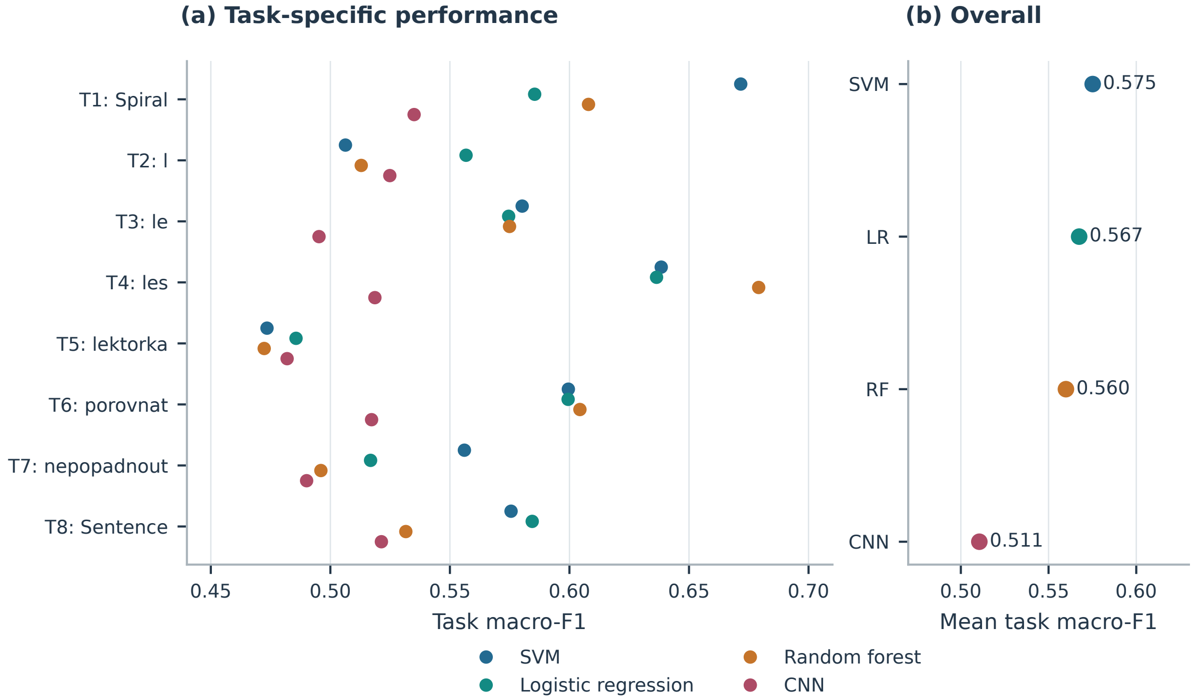

Figure 5: Classifier comparison on *SAZ*. Left: the task-specific results, each averaged over five repetitions. Right: the unweighted task mean. The classical methods use LPQ; the CNN uses pixels. Inferential contrasts concern the fixed predictions and this selected representation.

The fused SVM has PD sensitivity 0.7135 and H specificity 0.5895. Logistic regression has sensitivity 0.7081 and specificity 0.5947. Their error rates are not symmetric despite near-balanced class counts. The PD decision on tied votes contributes to the interpretation of this operating point; no alternative threshold was chosen from the outer labels. The vote fraction gives an SVM AUC of 0.6808 and a logistic-regression AUC of 0.6658, even though logistic regression ranks slightly higher by thresholded macro F1.

The larger fused scores describe a participant-level decision. We do not test their difference from task-mean scores as a paired improvement, because the two endpoints summarize different predictions.

7 WHAT DOES THE CNN USE?

7.1 Explanations of Held-Out Predictions

The CNN's weak classification result raises a specific question: do its predictions depend on the handwriting region, or on artefacts of image placement and background? We apply XAI after freezing all outer checkpoints. Each image is explained by the exact checkpoint that produced its held-out prediction, rather than by a model fitted later to the entire cohort.

Grad-CAM weights convolutional activation maps with gradients of a target score (Selvaraju et al., 2017). We target the originally predicted class and use the last ReLU before the fourth pooling layer. At this point the spatial map is 16×16 for a 128×128 input. For activation map A^k and predicted-class logit g_c, the attribution is

$$L_c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad \alpha_k^c = \frac{1}{UV} \sum_{u,v} \frac{\partial g_c}{\partial A_{uv}^k}. \quad (9)$$

For the single output logit g, the PD target is $g_c = g$ and the H target is $g_c = -g$. The positive map is interpolated to input size and divided by its maximum when nonzero. Heatmap colours express attribution magnitude, independently of the input's semantic channels.

We calculate Grad-CAM for all 2,985 held-out predictions. A separate occlusion analysis replaces each non-overlapping 16×16 input patch with white and measures the decrease in confidence for the original predicted class. Negative changes are clipped to zero for the spatial occlusion map. This analysis covers all 597 predictions in the first repetition.

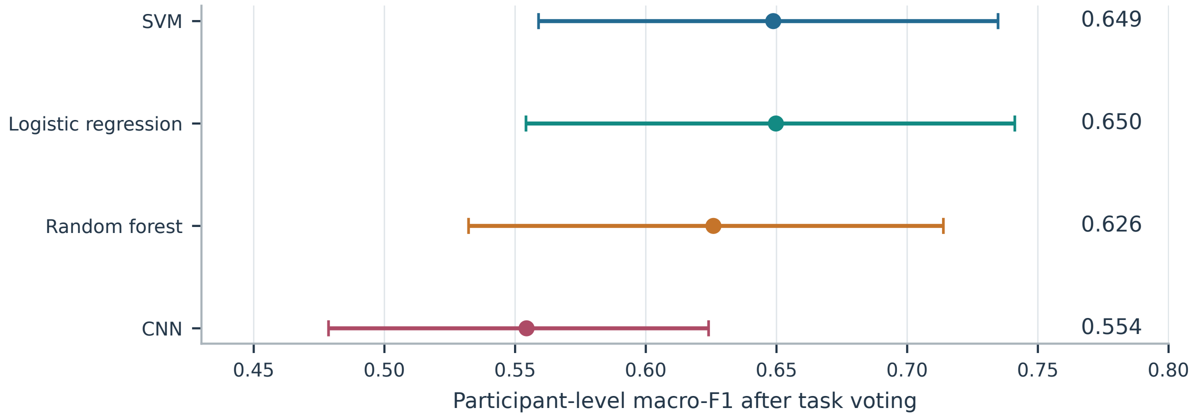

Figure 6: One prediction per participant obtained by voting over available tasks. Points average macro F1 over the five repetitions; intervals are conditional 95% participant-bootstrap intervals. These scores have a different prediction unit from the task means. None of the six paired fusion comparisons remains significant after Holm adjustment.

Figure 7 includes a correct and an incorrect prediction from each diagnosis class on T4. Examples were selected at median confidence within each stratum, before inspecting their explanations. The aggregate results cover every task; the four examples illustrate the methods rather than estimate their reliability.

7.2 Spatial Concentration and Perturbation

We define foreground as pixels with at least one channel below 250, matching the preprocessing. Foreground enrichment is the fraction of attribution inside that mask divided by the fraction of image area occupied by the mask. A uniform nonzero attribution map has enrichment one.

The participant-averaged Grad-CAM enrichment is 1.803, with interval [1.723, 1.884]. The result shows concentration around the rendered writing. It does not establish a spatial feature specific to PD: the foreground also indicates where any writing, from either class, is located. There are 174 all-zero Grad-CAM maps, 5.83% of the predictions. Enrichment is undefined for those maps, and the reported mean omits them before aggregation.

To test sensitivity to the highlighted pixels (Figure 8), we whiten the 10% of input pixels with the greatest Grad-CAM values. Confidence in the original class falls by 0.0489 on average, with interval [0.0427, 0.0553]. An equal-area random deletion gives a drop of 0.0033 [0.0016, 0.0050]. The paired advantage of the directed deletion is 0.0456 [0.0395, 0.0518]. Before bootstrapping, repeated predictions are averaged within each participant and task, and tasks are averaged within participant.

This control matches the number of replaced pixels. It does not match the amount of foreground removed or the spatial shape of the deleted region, and whitening can create inputs outside the training distribution. The larger directed effect is evidence of prediction sensitivity to the highlighted area, not a complete validation of attribution faithfulness.

Spatial agreement provides a less favourable result. The median pixelwise Spearman correlation between Grad-CAM and occlusion is -0.0121, using 540 of the 597 pairs with nonconstant maps. The remaining 57 correlations are undefined. Smooth gradient-based regions and patch perturbations do not give interchangeable accounts of the relevant locations in these images.

We also reinitialize all network weights and compare Grad-CAM for one deterministically chosen held-out image per checkpoint. The 200 checks cover only 11 distinct participants because the selection rule repeatedly chooses the first identifier in each test partition. The median trained–random correlation is 0.0276 among 160 nonconstant pairs. Low agreement is consistent with dependence on learned weights, but the limited participant coverage and excluded constant maps constrain this check.

7.3 Sensitivity to the Encoded Channels

A whole-channel perturbation addresses a different part of the representation: whether the CNN's confidence changes when one colour channel is replaced with white. Across the held-out predictions, the mean drops are 0.0429 for speed, 0.0482 for altitude and 0.0427 for azimuth. Their intervals are [0.0343, 0.0512], [0.0400, 0.0568] and [0.0360, 0.0494], respectively.

These similar global values conceal task varia-

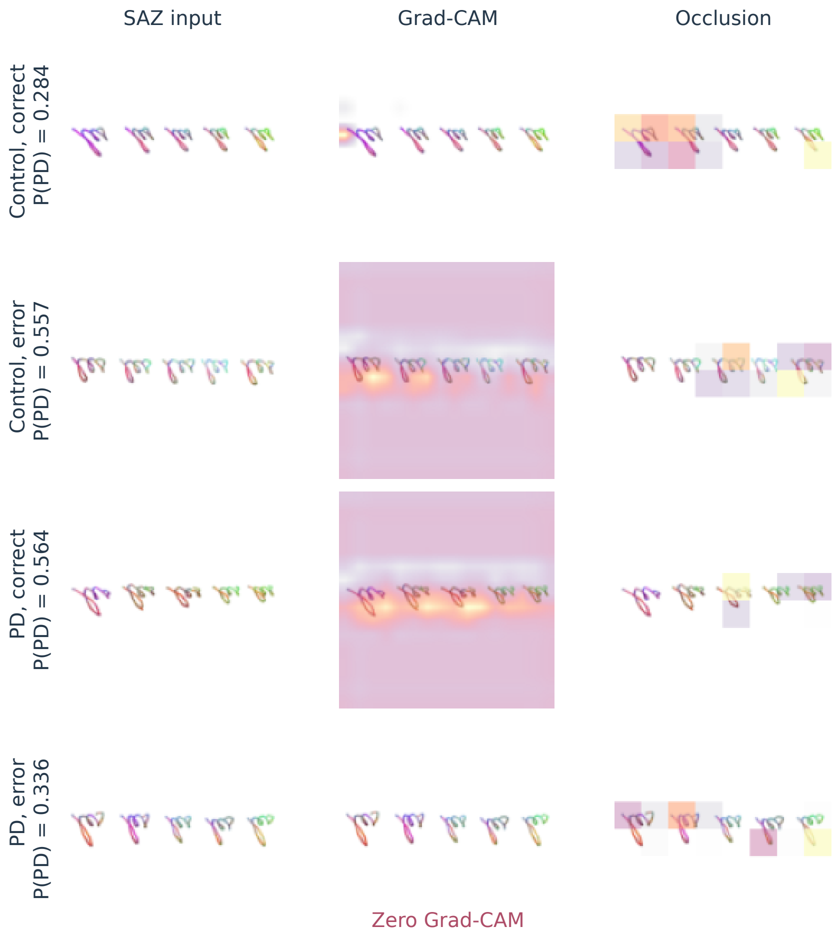

Figure 7: Task-4 explanations for correct and incorrect predictions from both classes. The four cases come from the first repetition, selected at median predicted-class confidence within each diagnosis/outcome stratum. Maps target the original predicted class and use its outer-test checkpoint. Opacity follows per-map attribution. The all-zero Grad-CAM in the PD error is retained. The CNN output is not a calibrated clinical probability.

tion. Speed has the largest average effect on the spiral, while its effect on T6 is slightly negative. Removing a channel alters colour contrast as well as its coded measurement. The perturbation therefore cannot rank the underlying physiological signals or settle the speed-versus-pressure choice in the earlier SVM experiment.

The XAI results describe this fitted CNN: gradients emphasize writing, but agree poorly with occlusion. Their clinical meaning cannot be established from a classifier with task-mean macro F1 of 0.5105.

8 DISCUSSION

8.1 What the Incremental Comparison Establishes

SAZ ranks first, but its advantage over static is about one macro-F1 percentage point and its interval includes zero. Normalized colour patterns summarized by LPQ did not provide a clear gain across all eight tasks. This finding concerns the representation and descriptor, not the general relevance of writing dynamics.

The renderer removes absolute size and signal levels, LPQ discards the location of each phase code, and the image omits in-air movement. These choices may discard useful information alongside nuisance variation. The present experiment cannot separate their effects.

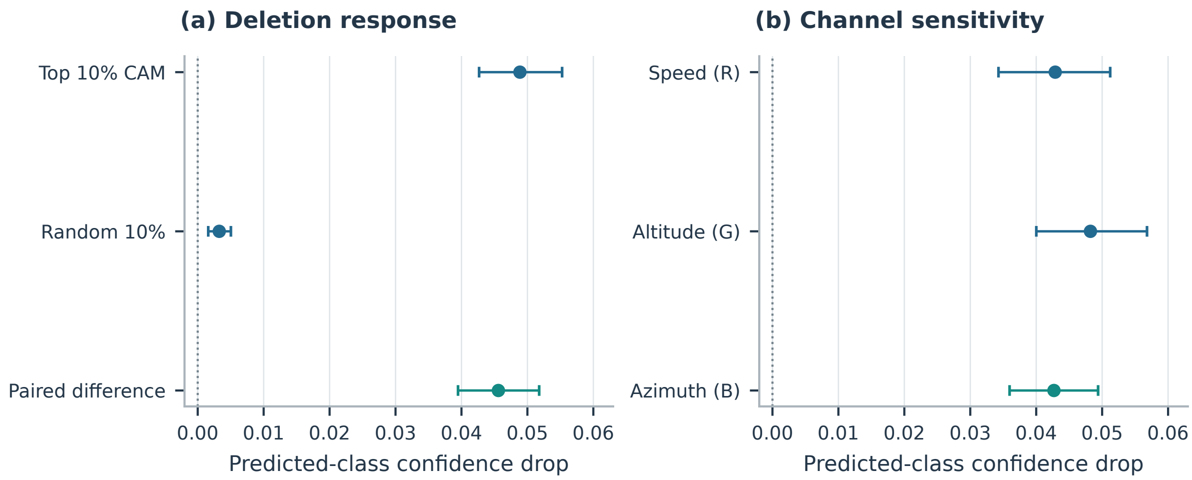

Figure 8: Confidence changes for the originally predicted class. Left: deleting the top 10% Grad-CAM pixels versus equal-area random deletion. Right: whitening each semantic channel. Intervals resample participants after averaging repeats and tasks. These perturbations measure sensitivity of the frozen CNN; the random control does not match foreground overlap, and channel removal also changes colour contrast.

Task dependence remains substantial: T4 is strongest for static SVM and *SAZ* Random Forest, whereas the spiral is strongest for *SAZ* SVM. The task average captures this variation, while fusion asks whether combining their predictions is useful for a participant-level decision.

8.2 Comparison with Handwriting Studies

Casademunt González (Casademunt González, 2023) evaluates CNNs on PaHaW crops of size 111 for T2, T3 and T4. The code separates 62 training and 13 test participants before producing crops. Its F1 concerns the positive PD class, rather than macro F1.

Table 3 compares reported accuracy with the task set identified. Casademunt's T2/T3/T4 accuracies exceed our *SAZ* SVM values of 51.20/58.13/64.00% and CNN values of 55.20/52.00/53.60%. This is not a paired comparison: the splits, number of training crops and selection procedures differ, and the thesis compares crop sizes using the designated test group.

Drotár et al. report 81.3% accuracy for kinematic and pressure features from T2–T8 (Drotár et al., 2016). Their descriptors can retain physical information removed by our normalization. They repeat tenfold validation ten times and screen features statistically; the description does not fully establish whether screening is nested in each split.

Diaz et al. report 86.67% accuracy for their enhanced representation and ensemble, versus 72.50% for its static counterpart (Diaz et al., 2019). The enhancement includes in-air movement, and their pipeline uses pretrained CNN features and five selected tasks. It preserves and models different information from our contact-only colour images with LPQ. These protocol differences preclude a controlled cross-paper ranking.

Our strongest participant-level result remains below these published accuracies. Different inputs and evaluation procedures may explain part of that gap, but do not make it disappear.

8.3 Limits of Complexity and Explanation

With approximately 60 outer-training participants, a single task offers little independent data for learning filters from scratch. Affine augmentation adds views, not people. The tested CNN's lower scores favour the fixed descriptor in this setting, without establishing that CNNs generally are unsuitable for handwriting.

An attribution can identify the strip occupied by writing without identifying stable disease-related features. The near-zero Grad-CAM–occlusion correlation exposes that limitation. Neither method explains why the CNN underperformed; that would require a separate controlled investigation.

Independent-cohort validation is needed to estimate generalization beyond this development process, including age matching and acquisition differences. Another random split of PaHaW would not undo its role in design decisions. Future work could select representations within training partitions or freeze them before collecting a new cohort.

Table 3: Contextual comparison on PaHaW. All values in this table are accuracy, not macro F1. Published results are transcribed from the cited works; none was rerun under our partitions. Different task sets and selection procedures prevent a direct superiority test.

Study	Input and classifier	Evaluation scope	Accuracy (%)
Casademunt González (2023)	CNN, handwriting crops	T2 / T3 / T4; 62 train, 13 test	69.23 / 61.54 / 69.23
Drotár et al. (2016)	Kinematics + pressure, SVM	T2–T8; repeated ten-fold CV	81.30
Diaz et al. (2019)	Enhanced images and task ensemble	Ten-fold CV; five selected tasks	86.67
Present experiment	*SAZ* + LPQ + SVM	T1–T8; 5×5 participant CV; majority fusion	65.07
Present experiment	*SAZ* + LPQ + logistic regression	T1–T8; 5×5 participant CV; majority fusion	65.07

8.4 Reproducibility and Scope

Saved configurations, participant assignments, rendering diagnostics, inner searches and outer predictions support reproduction. XAI reproduces checkpoint probabilities before explanation, and checksums link figure inputs and tables to source files.

The scope is fixed-thickness, contact-only colour encoding. Absolute magnitude and spatial location can both matter for classification; the representation and descriptor make explicit compromises on each.

9 CONCLUSION

Across eight PaHaW tasks, explicit RGB combinations changed performance only modestly: speed, altitude and azimuth ranked first, without a supported advantage over the static reconstruction. On that representation, LPQ with SVM or logistic regression was more effective than the small CNN trained from scratch. Combining task predictions gave the strongest participant-level scores, with the two classical methods near 0.65 macro F1.

Grad-CAM concentrated on the handwriting and guided stronger deletions than an equal-area random control, yet agreed poorly with occlusion. These explanations show the CNN's sensitivity to image regions without establishing disease-specific features. Under the tested normalization and participant partitions, neither additional colour signals nor the CNN trained from scratch produced a clear advantage over the corresponding simpler alternatives.

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